

Loan Approval Analysis



Introduction

In this assessment, you will perform an Exploratory Data Analysis (EDA) on a dataset related to home loan approval. The dataset is sourced from Skill Circle and contains valuable information for analysis. The primary focus of this assessment is on data exploration and visualization machine learning techniques are not required. The goals of this assessment are to: Gain familiarity with the dataset. Perform data exploration and visualization. Identify patterns, trends, and potential insights. Generate meaningful visualizations to communicate your findings.

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Libraries used: -

1. Data Manipulation and Cleaning

Pandas: For handling and manipulating data in Data Frames, handling missing values, and performing group-wise operations.

NumPy: For numerical operations, array manipulations, and mathematical computations.

2. Data Visualization

Matplotlib: To create static, animated, and interactive visualizations such as bar charts, scatter plots, and histograms.

Seaborn: For advanced statistical visualizations like box plots, heatmaps, and pair plots.

Plotly : For interactive and dynamic visualizations, like zoomable scatter plots and clickable heatmaps.

IMPORT AND LOAD DATASET

```
[ ] import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
import seaborn as sns
```

```
# Load Dataset
data=pd.read_csv('loan_sanction_test.csv')
data
```

	Loan_ID	Gender	Married	Dependents	Education	Self_Employed	ApplicantIncome	CoapplicantIncome	LoanAmount	Loan_Amount_Term	Credit_History	Property_Area
0	LP001015	Male	Yes	0	Graduate	No	5720	0	110.0	360.0	1.0	Urban
1	LP001022	Male	Yes	1	Graduate	No	3076	1500	126.0	360.0	1.0	Urban
2	LP001031	Male	Yes	2	Graduate	No	5000	1800	208.0	360.0	1.0	Urban
3	LP001035	Male	Yes	2	Graduate	No	2340	2546	100.0	360.0	NaN	Urban
4	LP001051	Male	No	0	Not Graduate	No	3276	0	78.0	360.0	1.0	Urban
...	...	...	...	...	...	...	...	...	...	...	...	...
362	LP002971	Male	Yes	3+	Not Graduate	Yes	4009	1777	113.0	360.0	1.0	Urban
363	LP002975	Male	Yes	0	Graduate	No	4158	709	115.0	360.0	1.0	Urban
364	LP002980	Male	No	0	Graduate	No	3250	1993	126.0	360.0	NaN	Semiurban
365	LP002986	Male	Yes	0	Graduate	No	5000	2393	158.0	360.0	1.0	Rural
366	LP002989	Male	No	0	Graduate	Yes	9200	0	98.0	180.0	1.0	Rural

267 rows x 13 columns

Overview and feature description

Understanding each feature in the dataset is essential for effective EDA. The features in this loan dataset can be categorized into applicant demographics, financial attributes, and loan-specific details:

Gender – Male or Female.

Married – Marital status of the applicant.

Dependents – Number of dependents (0, 1, 2, 3+).

Education – Graduate or Not Graduate.

Self_Employed – Employment type.

Applicant Income – Monthly income of the applicant.

Coapplicant Income – Income of coapplicant.

Loan Amount – Total loan amount requested.

Loan_Amount_Term – Duration of the loan in months.

Credit_History – Credit score status (1 = good history, 0 = no/poor history).Property_Area – Urban, Semiurban, or Rural

Handling missing values

Missing Value Analysis Missing value analysis is an essential step in any data cleaning process, as it identifies gaps that could distort results or impede analysis. In the loan dataset, several columns contain missing entries, which could be due to incomplete application forms or system recording issues.

Commonly affected features include **Loan Amount**, **Credit_History**, **Self_Employed**, and **Gender**.



Missing values in the dataset:

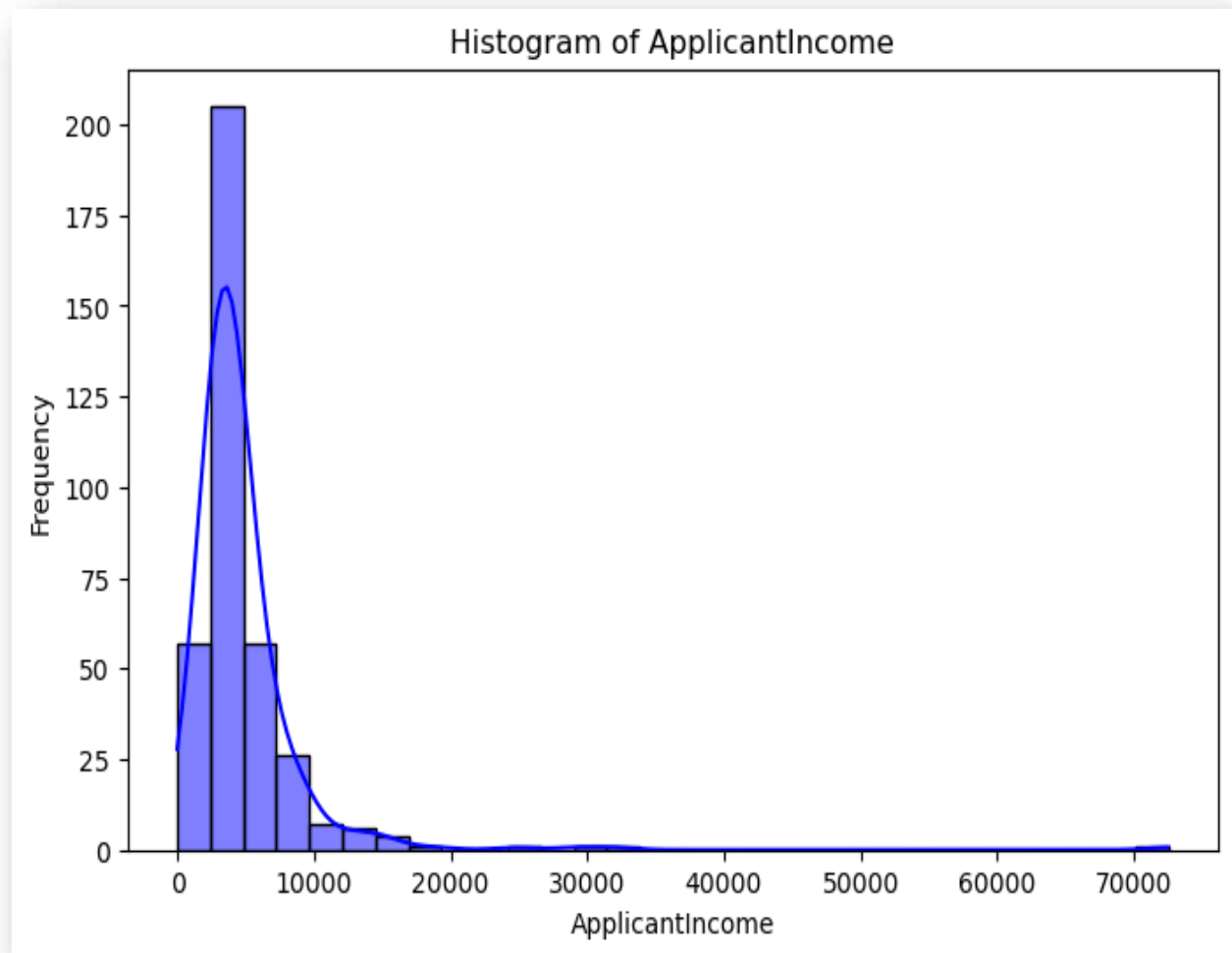
Loan_ID	0
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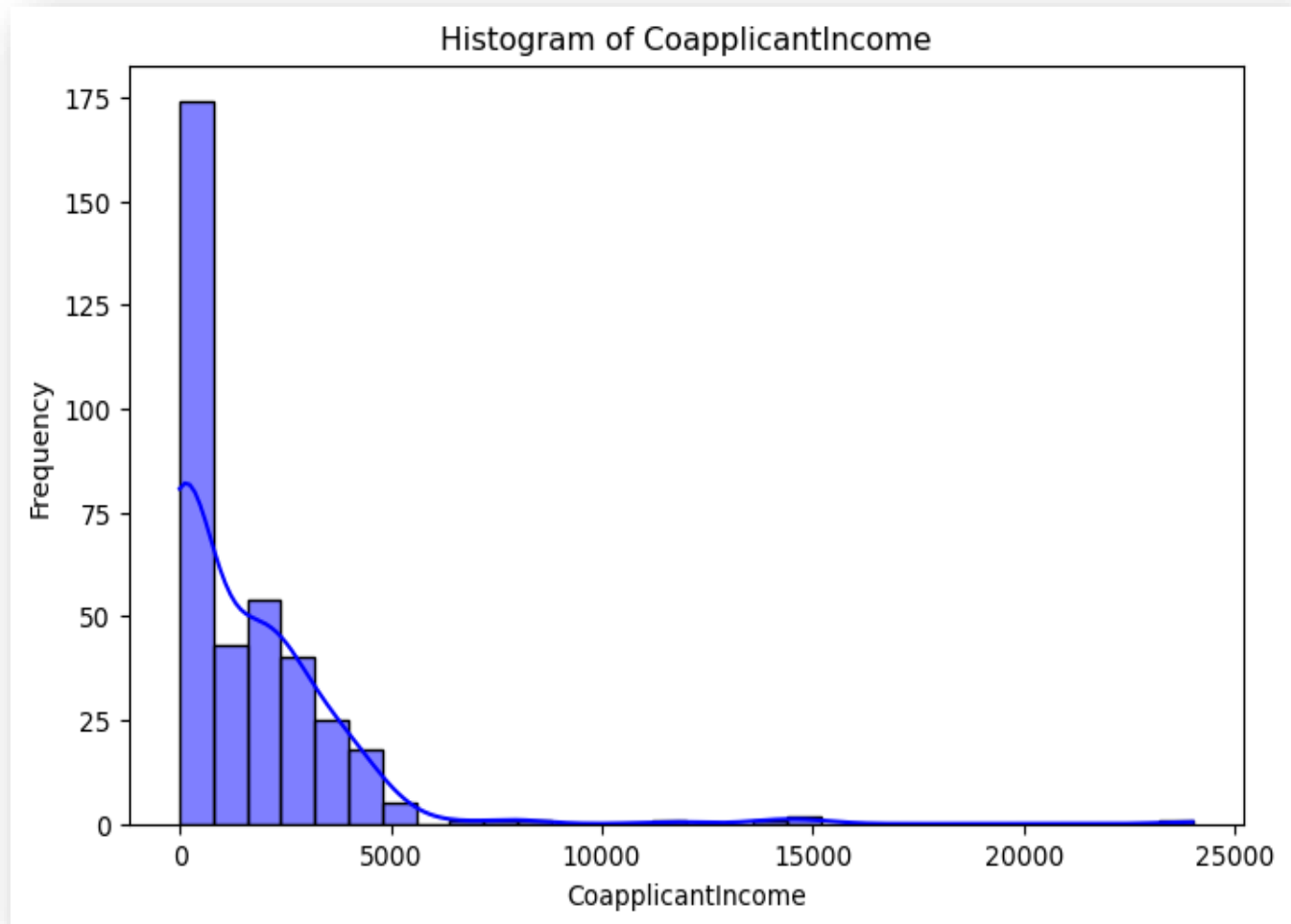
Gender	11
Married	0
Dependents	10
Education	0
Self_Employed	23
ApplicantIncome	0
Loan Amount	5
Loan_Amount_Term	6
Credit_History	29
Property_Area	0
dtype: int64	

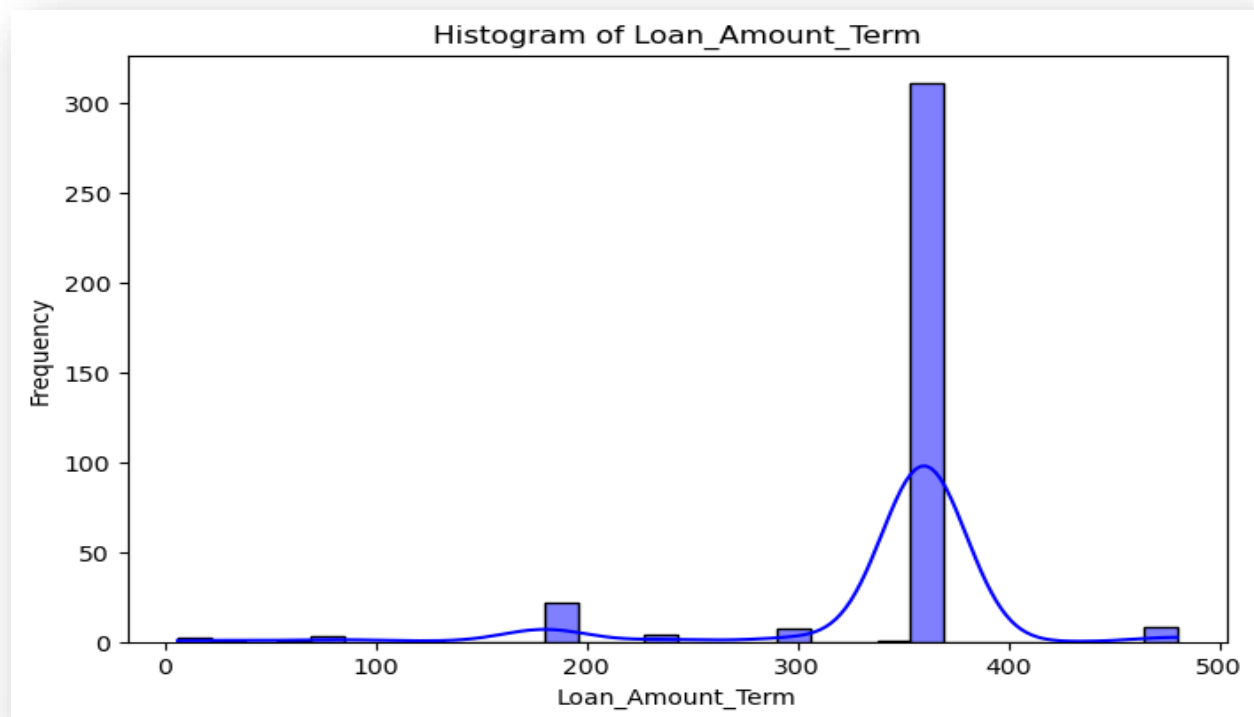
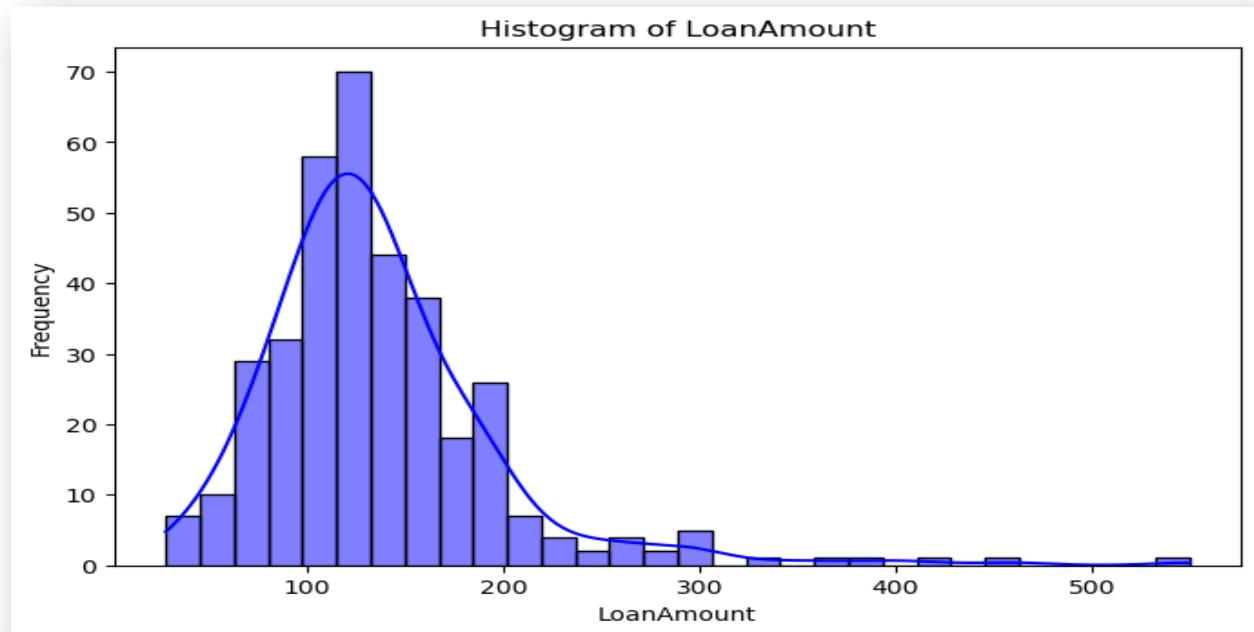
Univariate Analysis

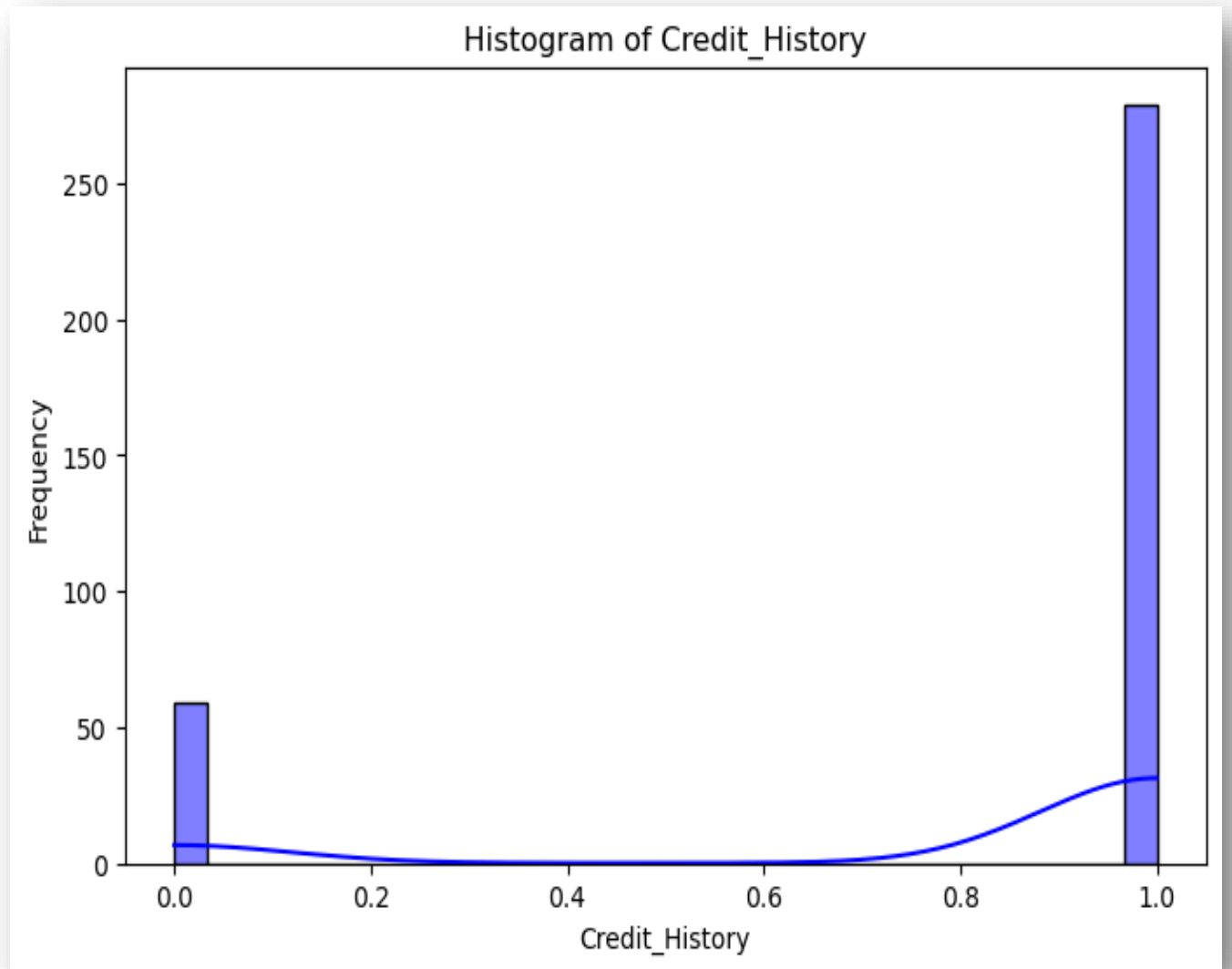
distribution of numeric columns using the following visualizations:

Through histogram



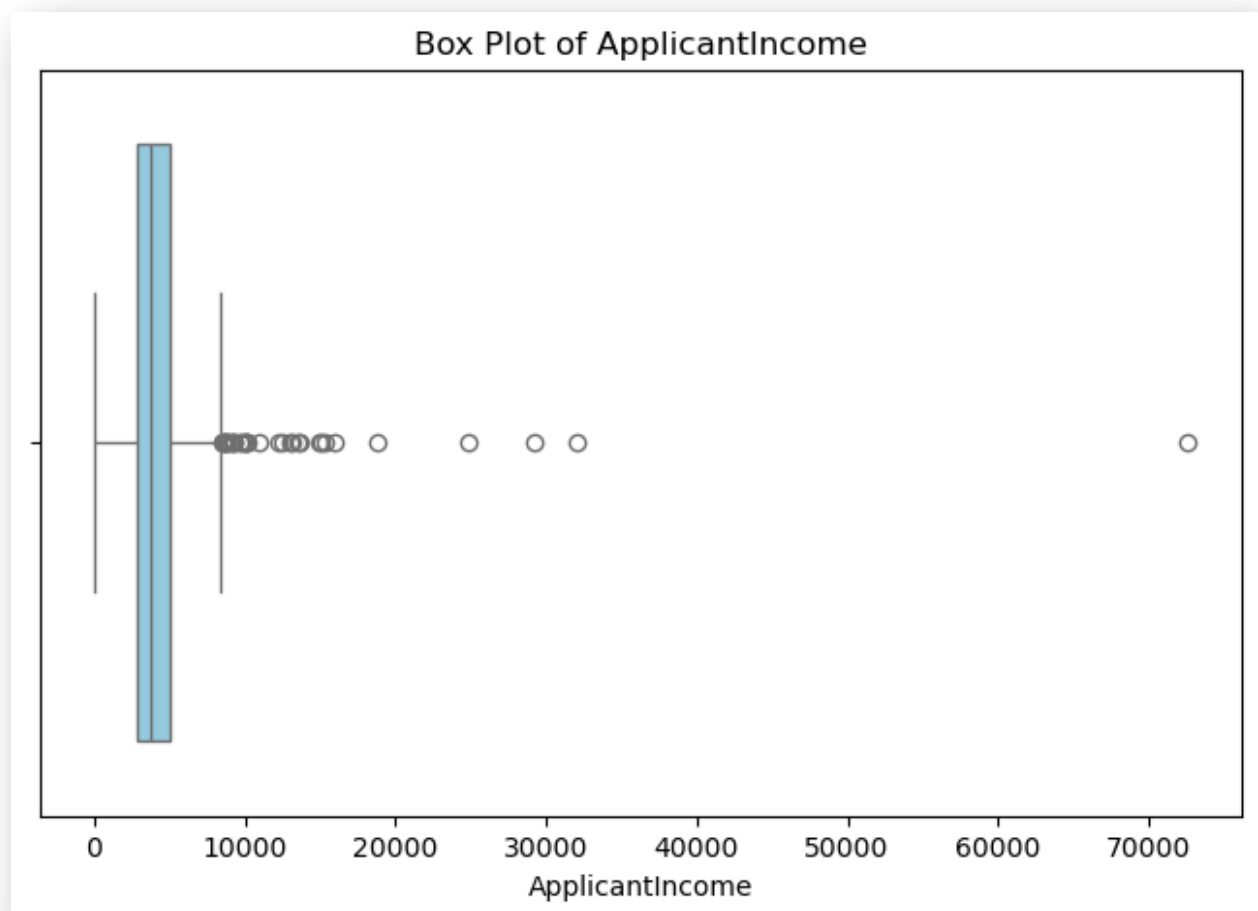


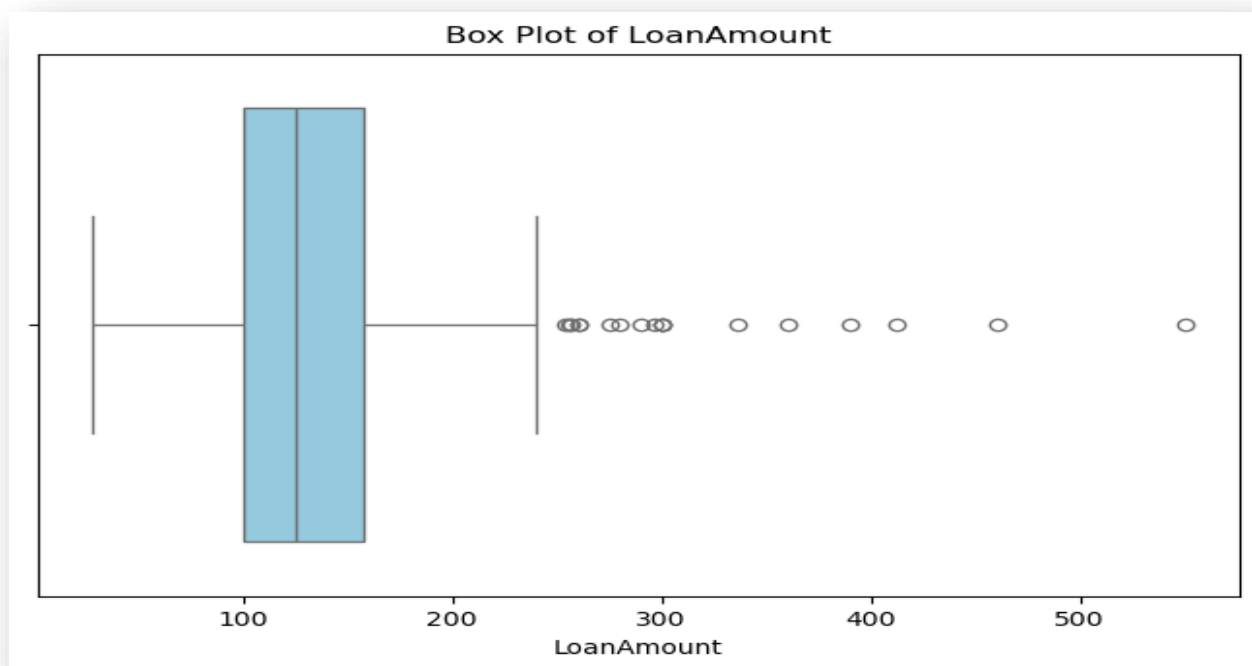
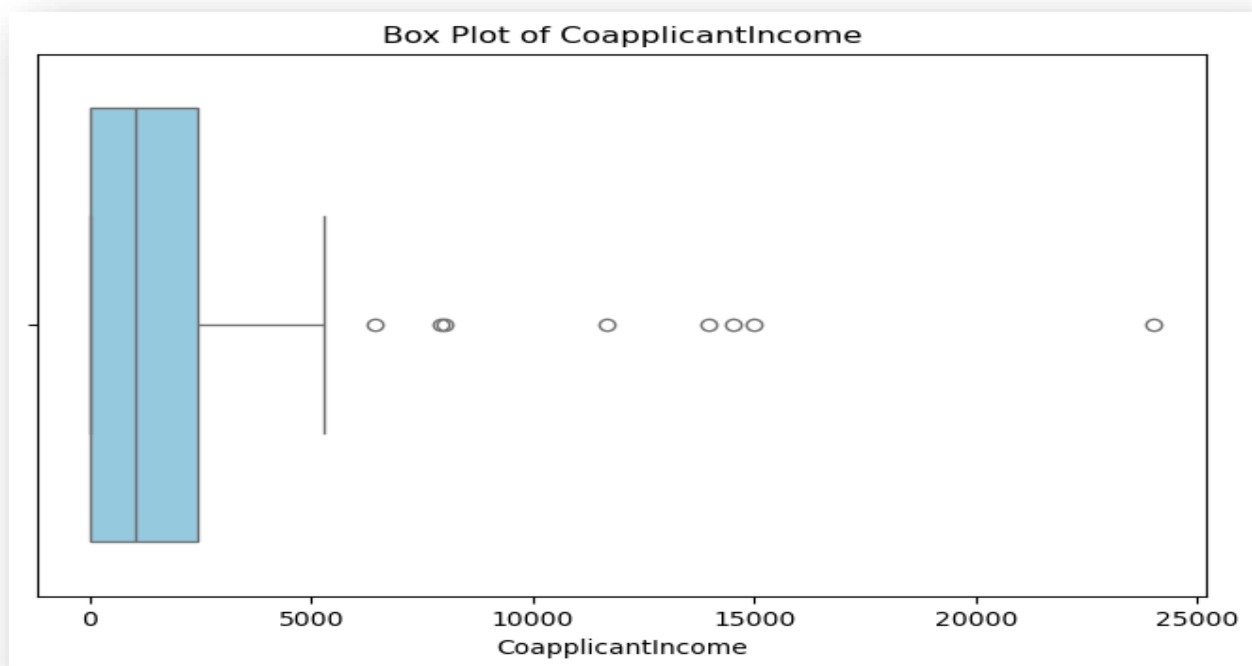


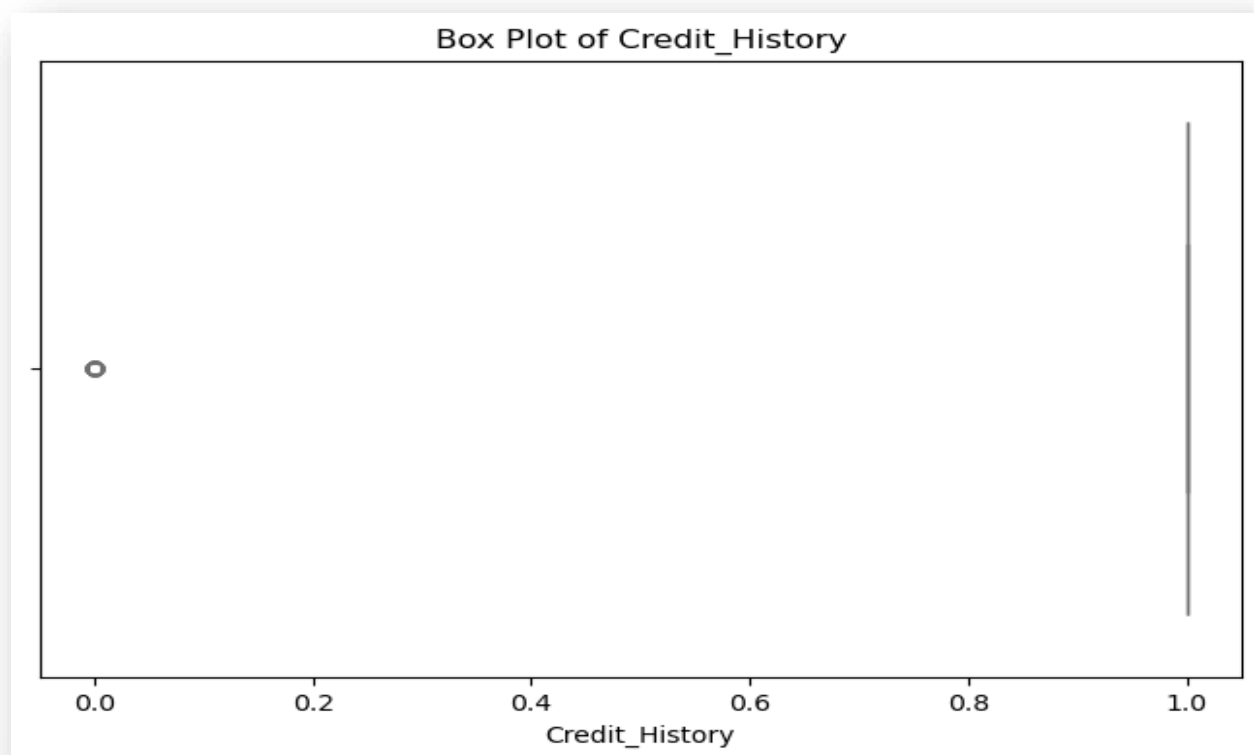
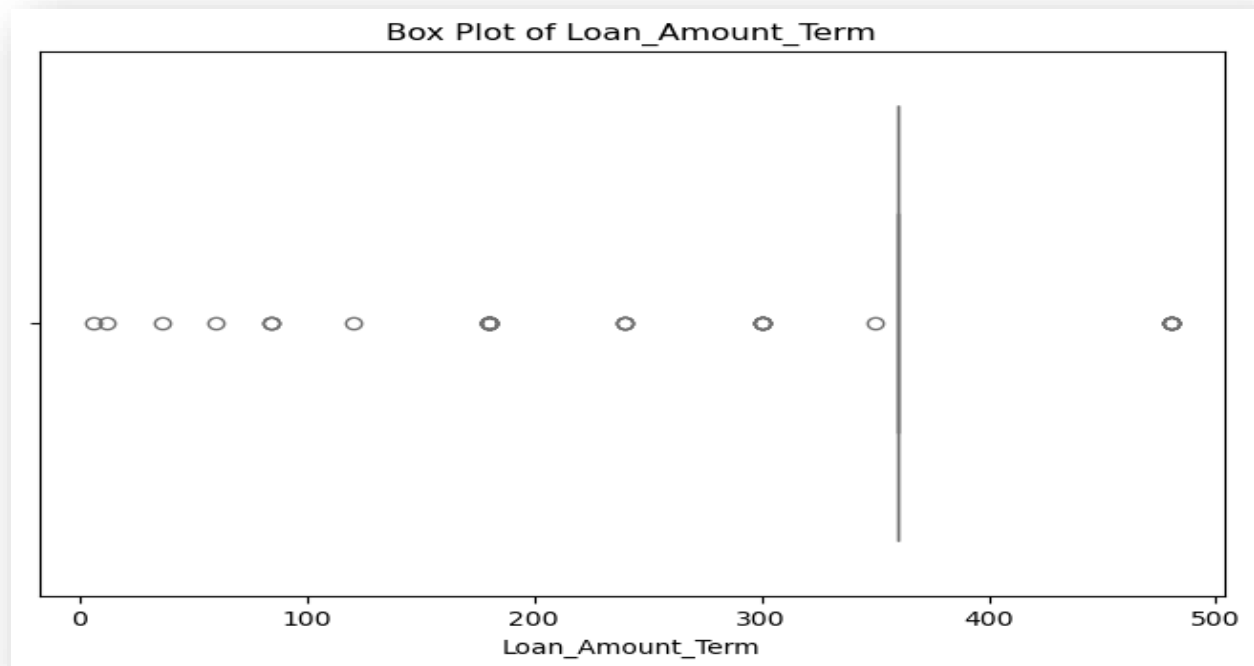


Through Box Plots: Identify potential outliers and visualize the spread of data.

```
for col in numeric_columns:
    plt.figure(figsize=(8, 5))
    sns.boxplot(x=data[col], color='skyblue')
    plt.title(f'Box Plot of {col}')
    plt.xlabel(col)
    plt.show()
```

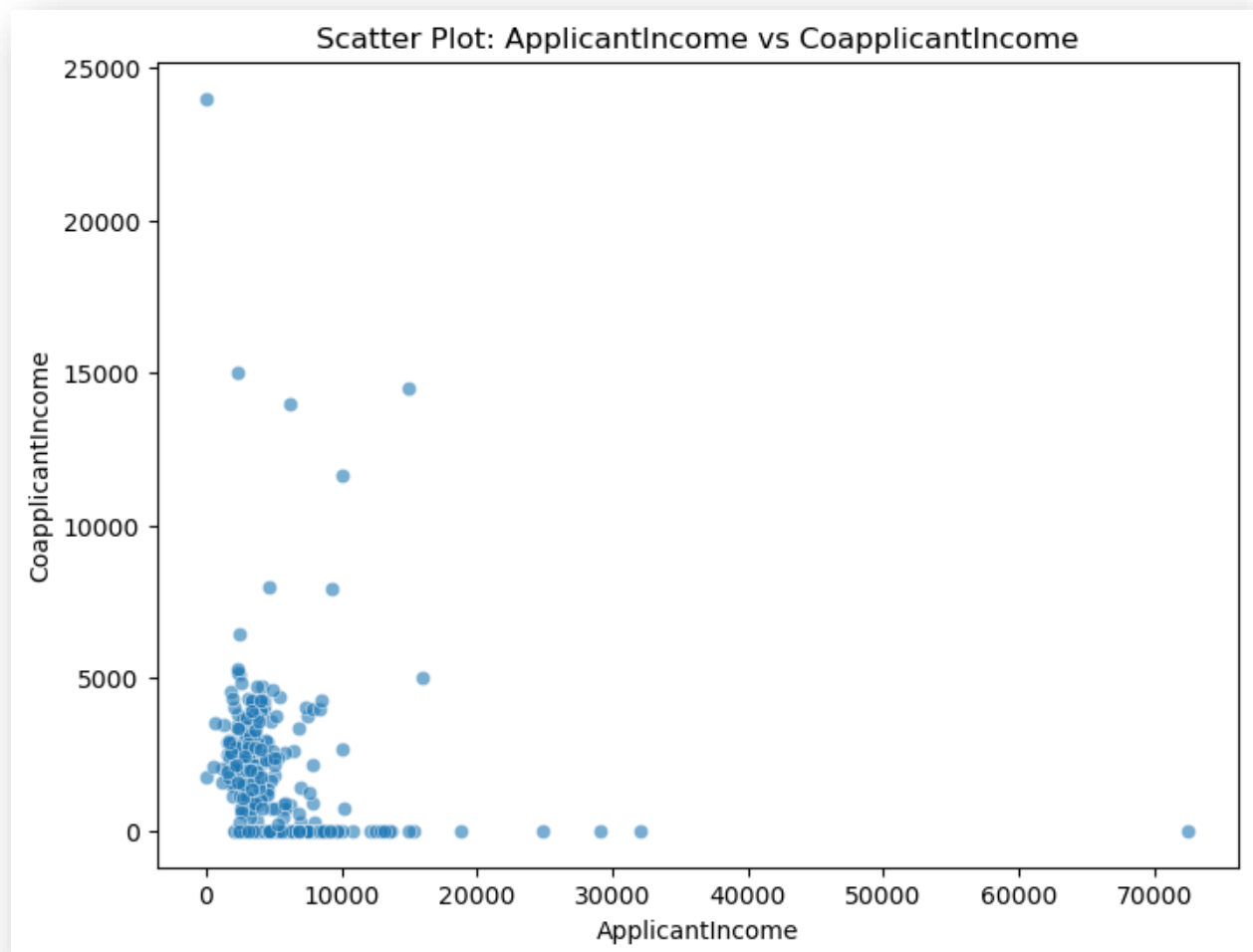


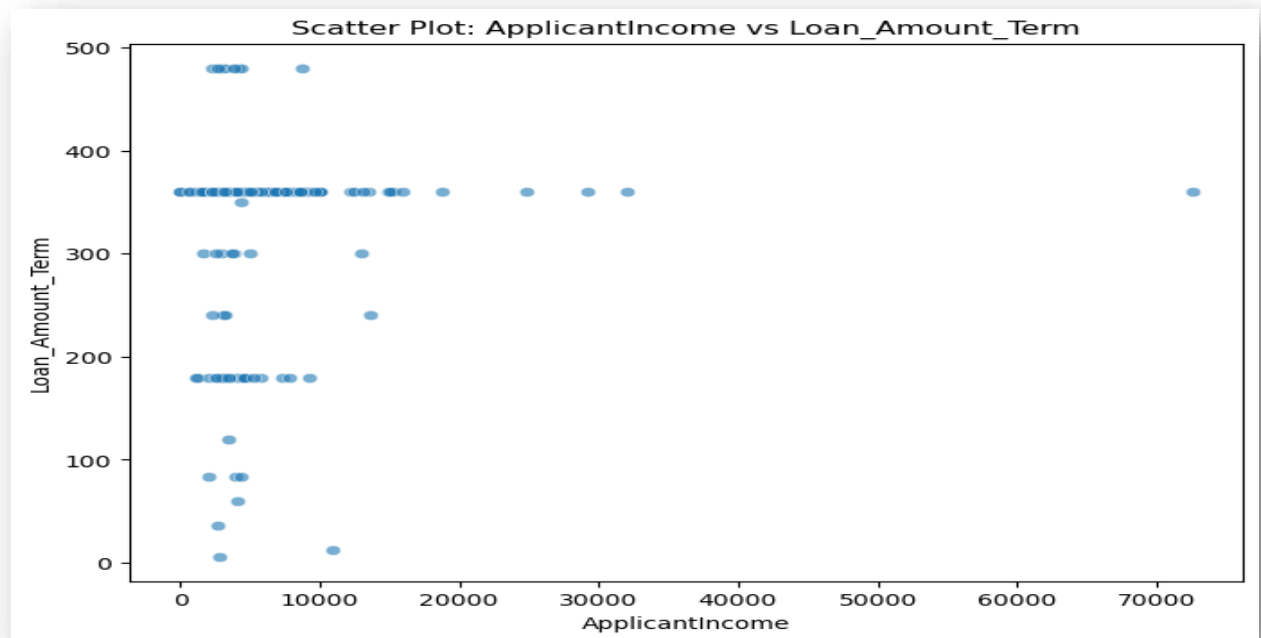
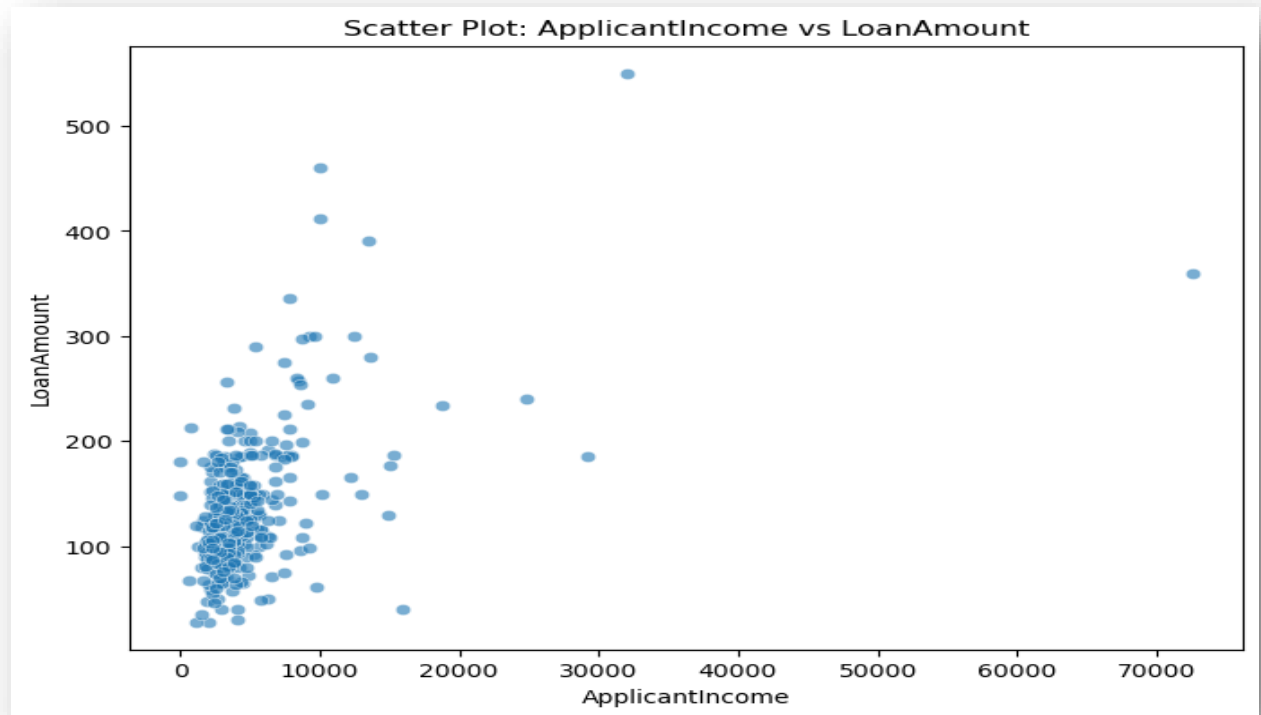


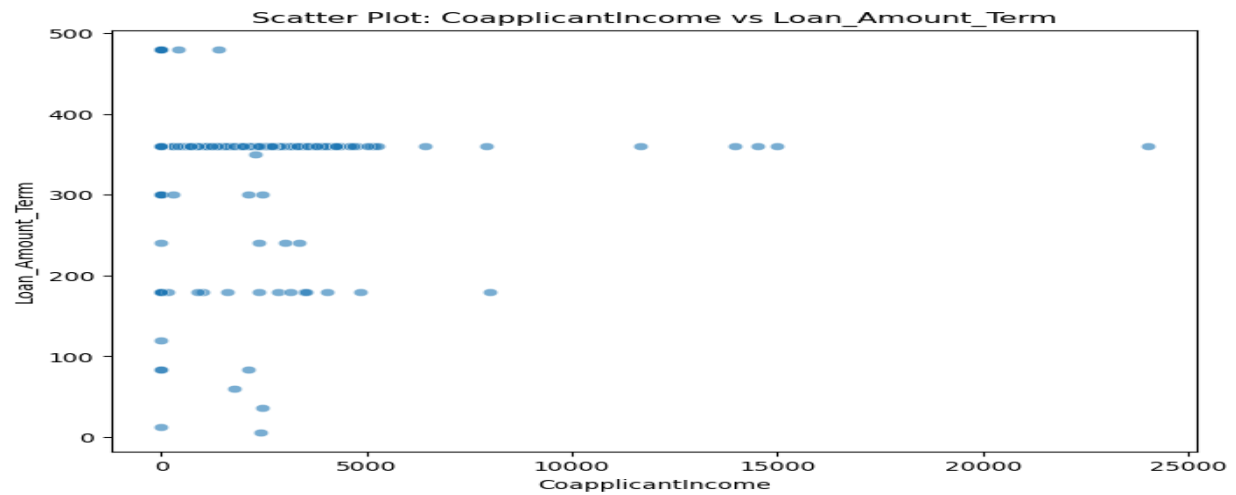
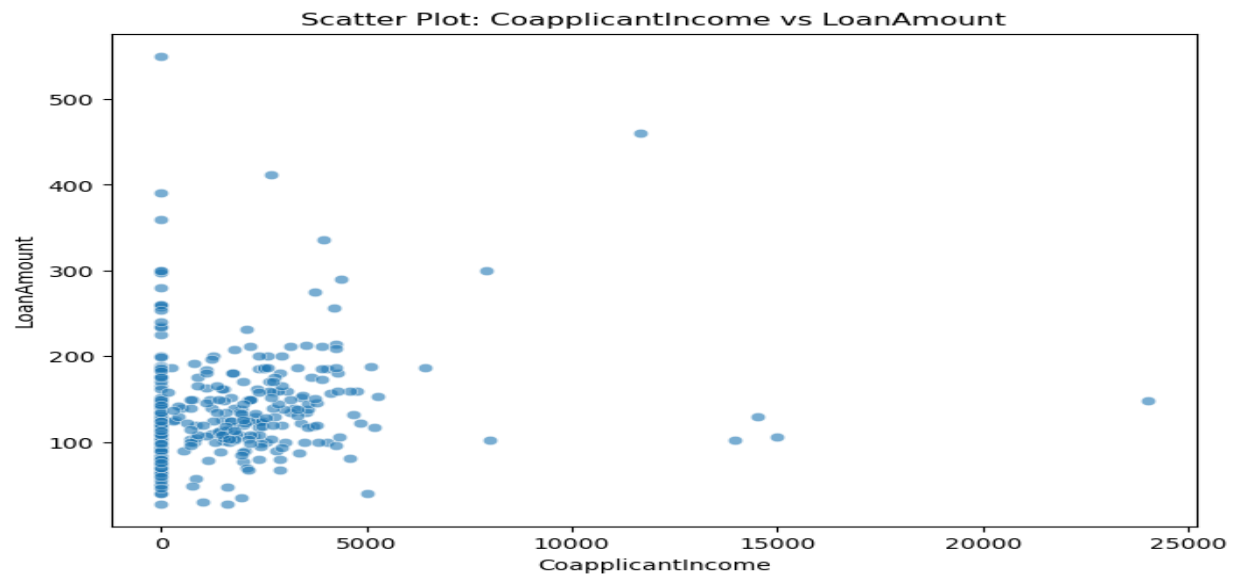
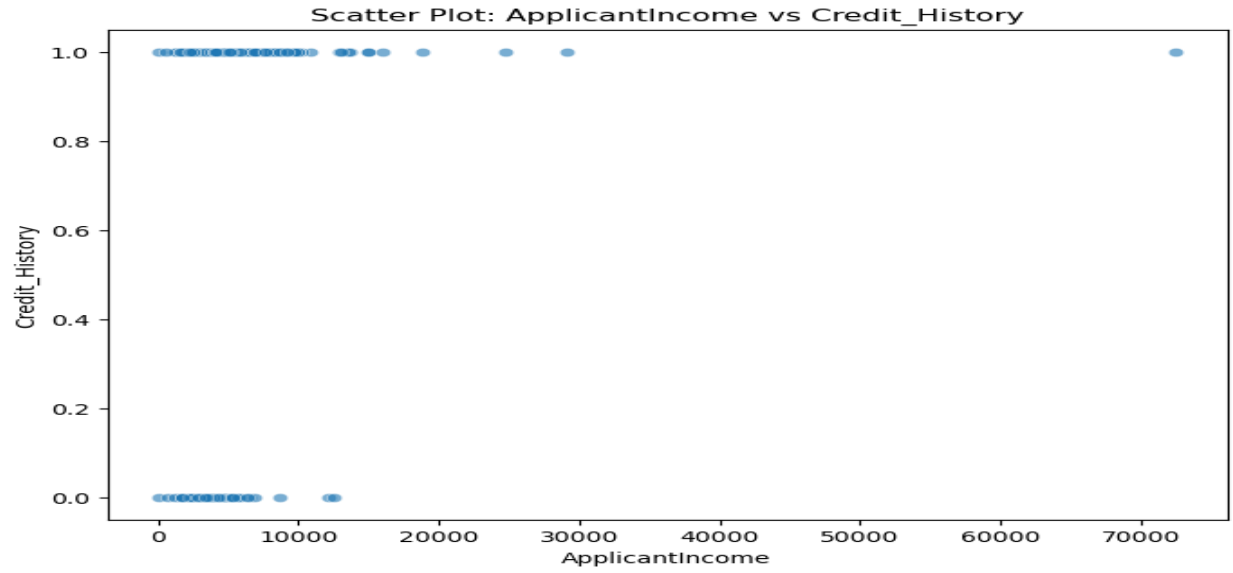


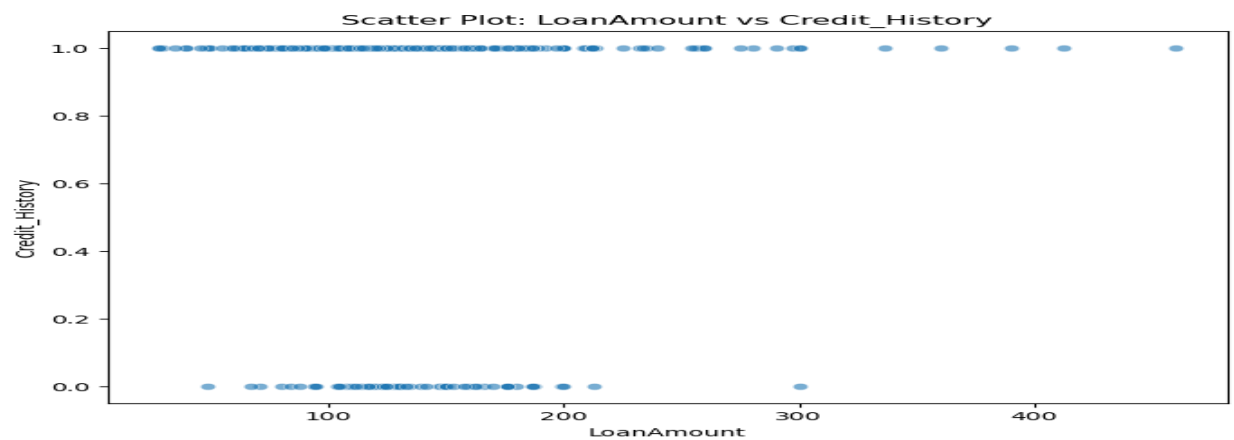
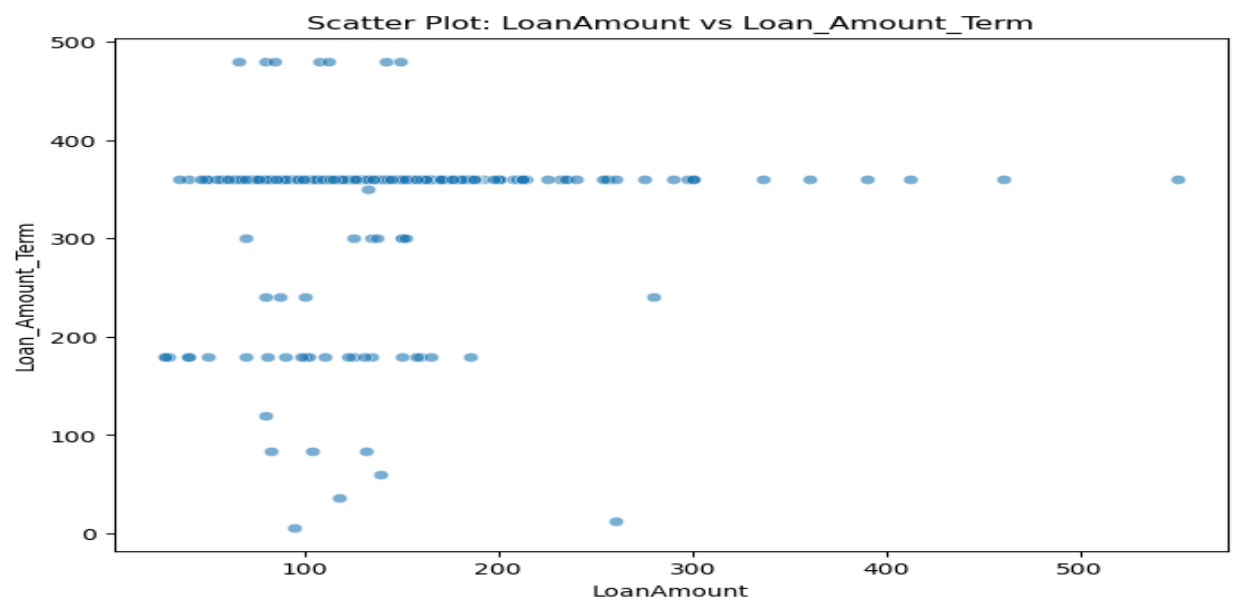
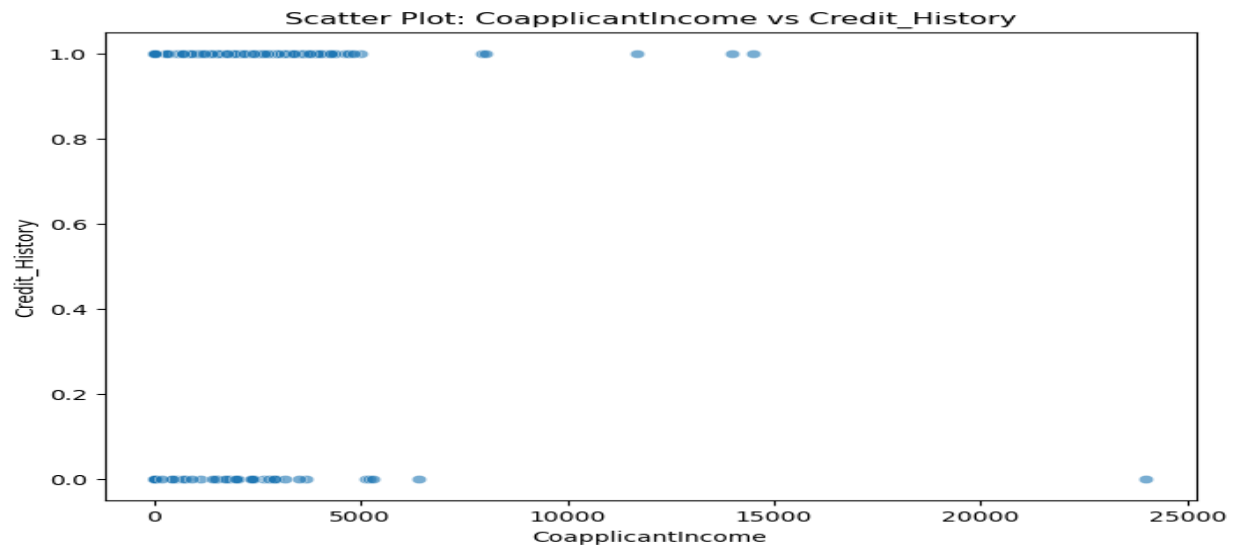
Bivariate analysis

scatter plots to explore relationships between pairs of numeric variables.









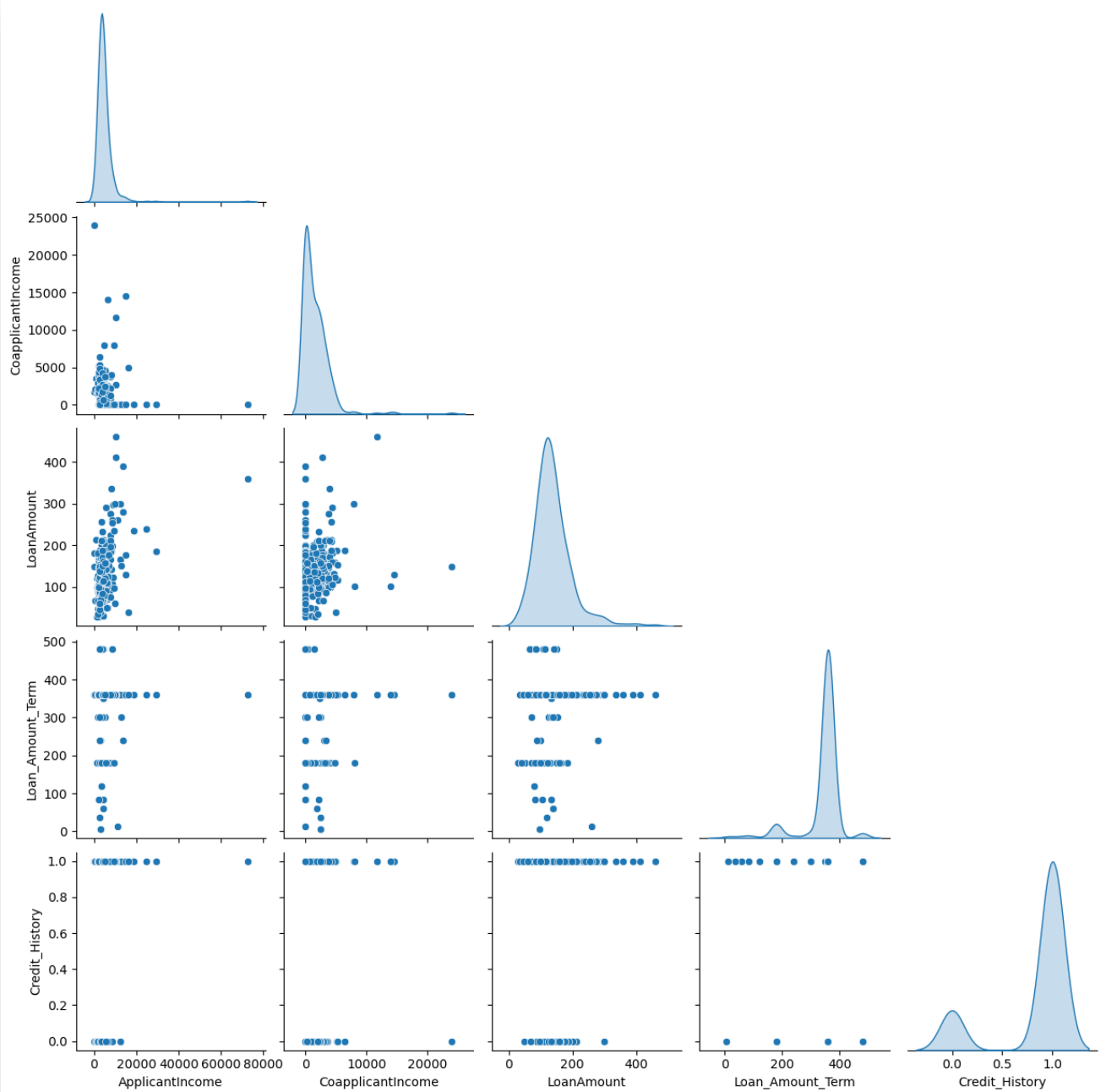
Some insights

Bivariate analysis is key for understanding the relationships between two variables. For loan approval, exploring the connection between numeric variables such as **ApplicantIncome**, **LoanAmount**, and **Loan_Amount_Term** with the **Loan_Status** helps identify patterns that influence loan decisions.

One of the most effective ways to visualize these relationships is through scatter plots. For example, a scatter plot can be used to examine the relationship between **ApplicantIncome** and **LoanAmount**. A positive correlation might suggest that higher-income applicants tend to apply for larger loans, which could be expected. Additionally, plotting **LoanAmount** against **Loan_Amount_Term** might reveal whether applicants with larger loan amounts generally request longer terms. This insight could be useful for assessing lending risk and repayment capacity.

Conclusion from Bivariate Analysis: From these scatter plots, we observe a positive correlation between applicant income and loan amount, suggesting that higher-income individuals often request more substantial loans. The relationship between income and coapplicant income further highlights that joint applicants tend to have a higher likelihood of loan approval, as lenders see lower risk

Through pairplot: Use pair plots (scatter matrix) to visualize interactions between multiple numeric variables simultaneously.



Investigate the relationship: between categorical and numeric variables using box plots or violin plots.

```

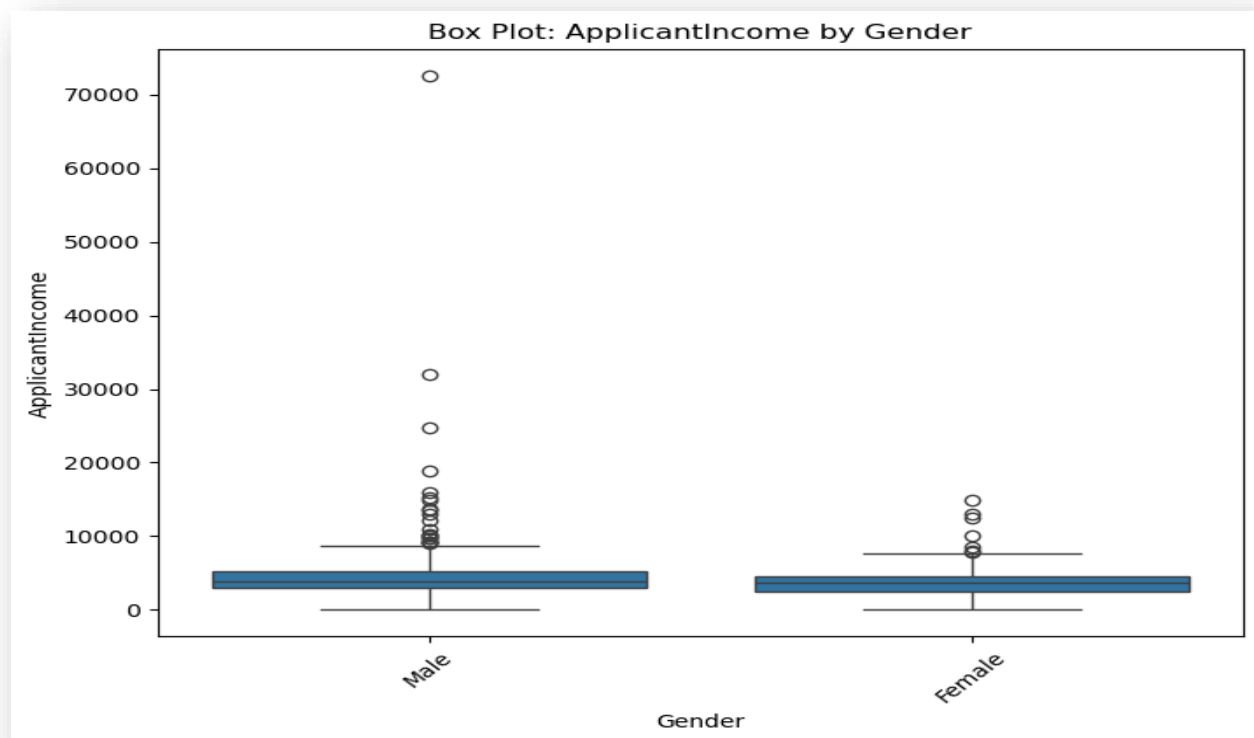
categorical_columns = ['Gender', 'Married', 'Dependents', 'Education',
'Self_Employed', 'Property_Area']
for cat_col in categorical_columns:

```

```

for num_col in numeric_columns:
    plt.figure(figsize=(8, 6))
    sns.boxplot(x=cat_col, y=num_col, data=data)
    plt.title(f'Box Plot: {num_col} by {cat_col}')
    plt.xlabel(cat_col)
    plt.ylabel(num_col)
    plt.xticks(rotation=45)
    plt.show()

```

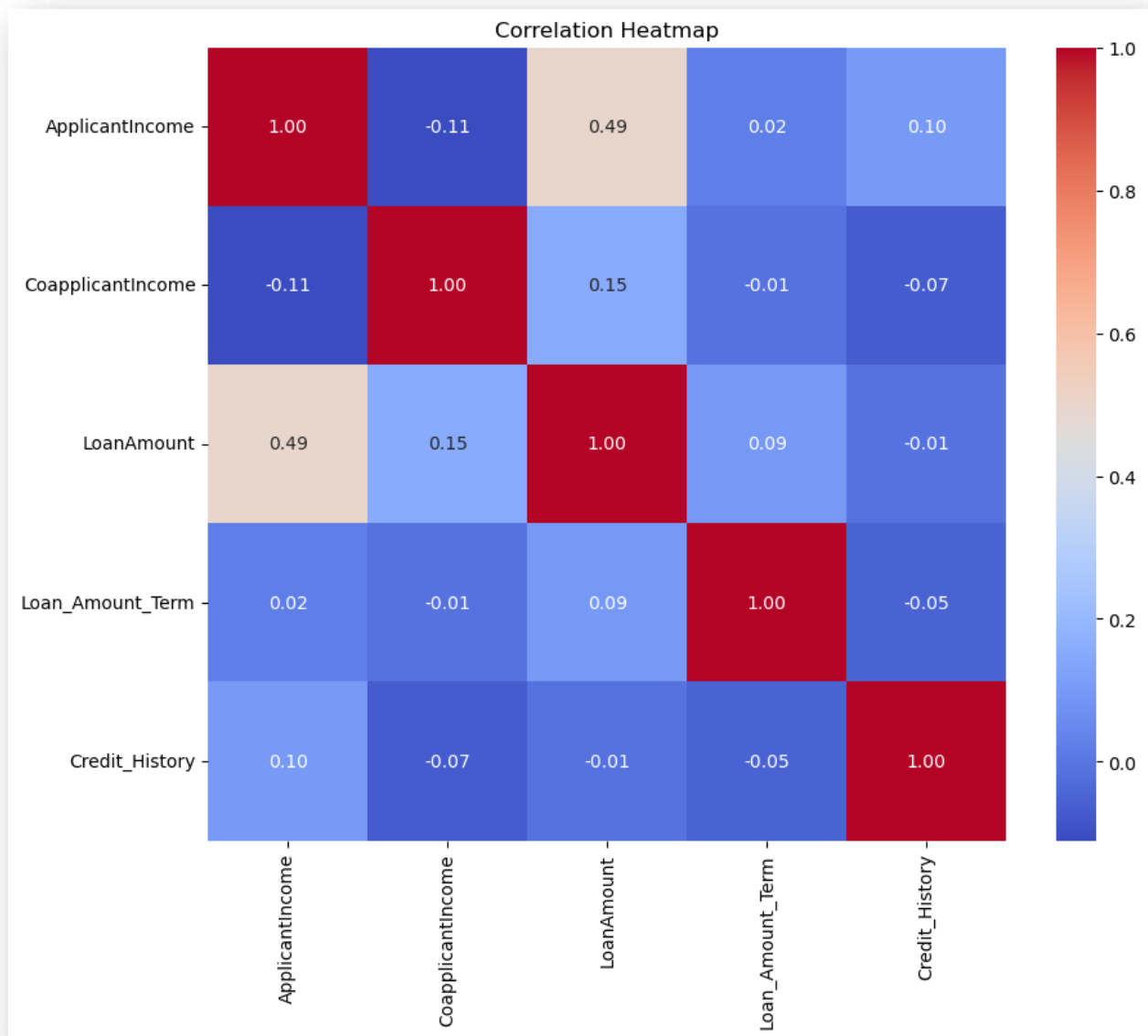


+30 similar shown in the .ipynb file link in the last page

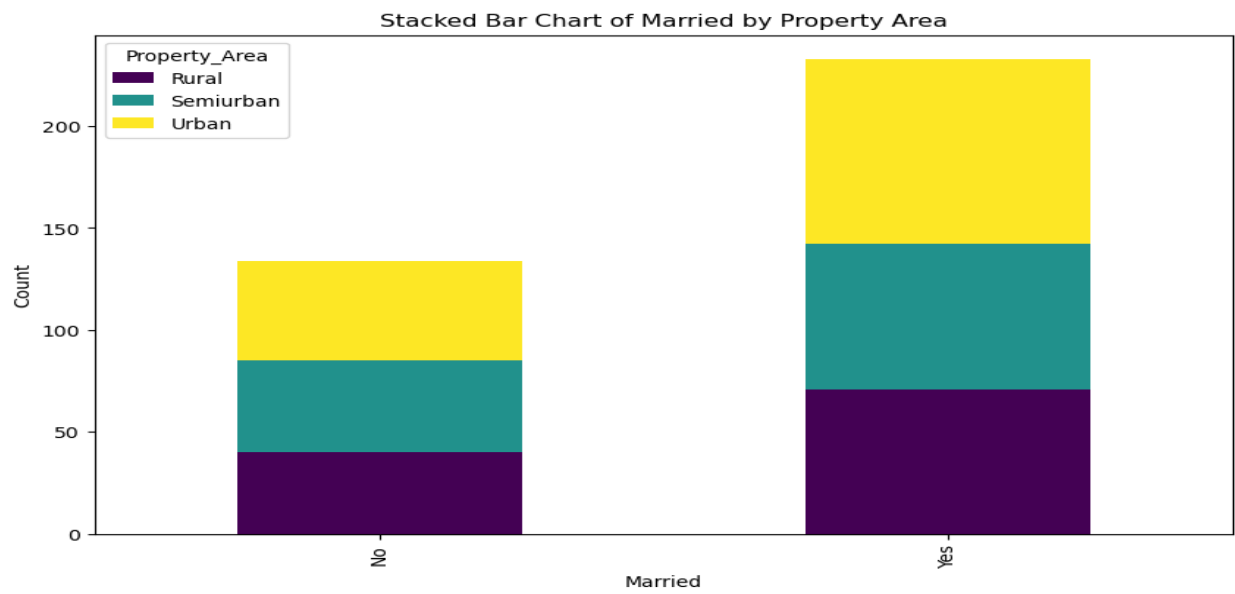
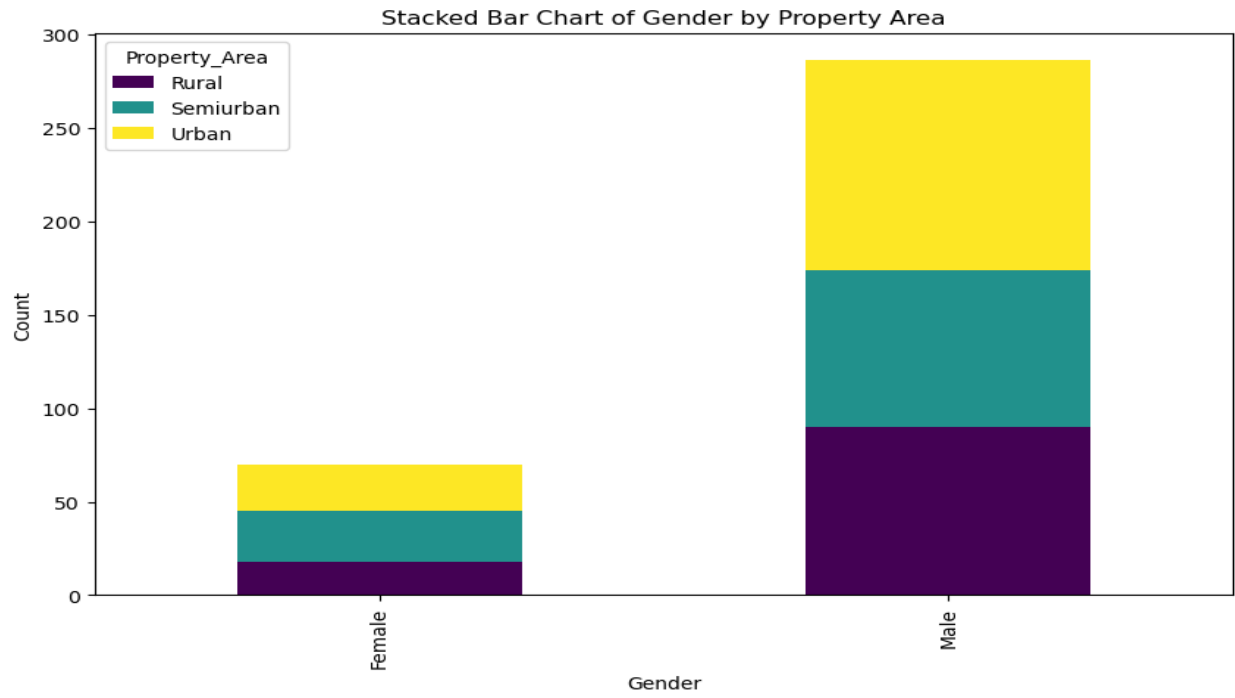
Multivariate Analysis

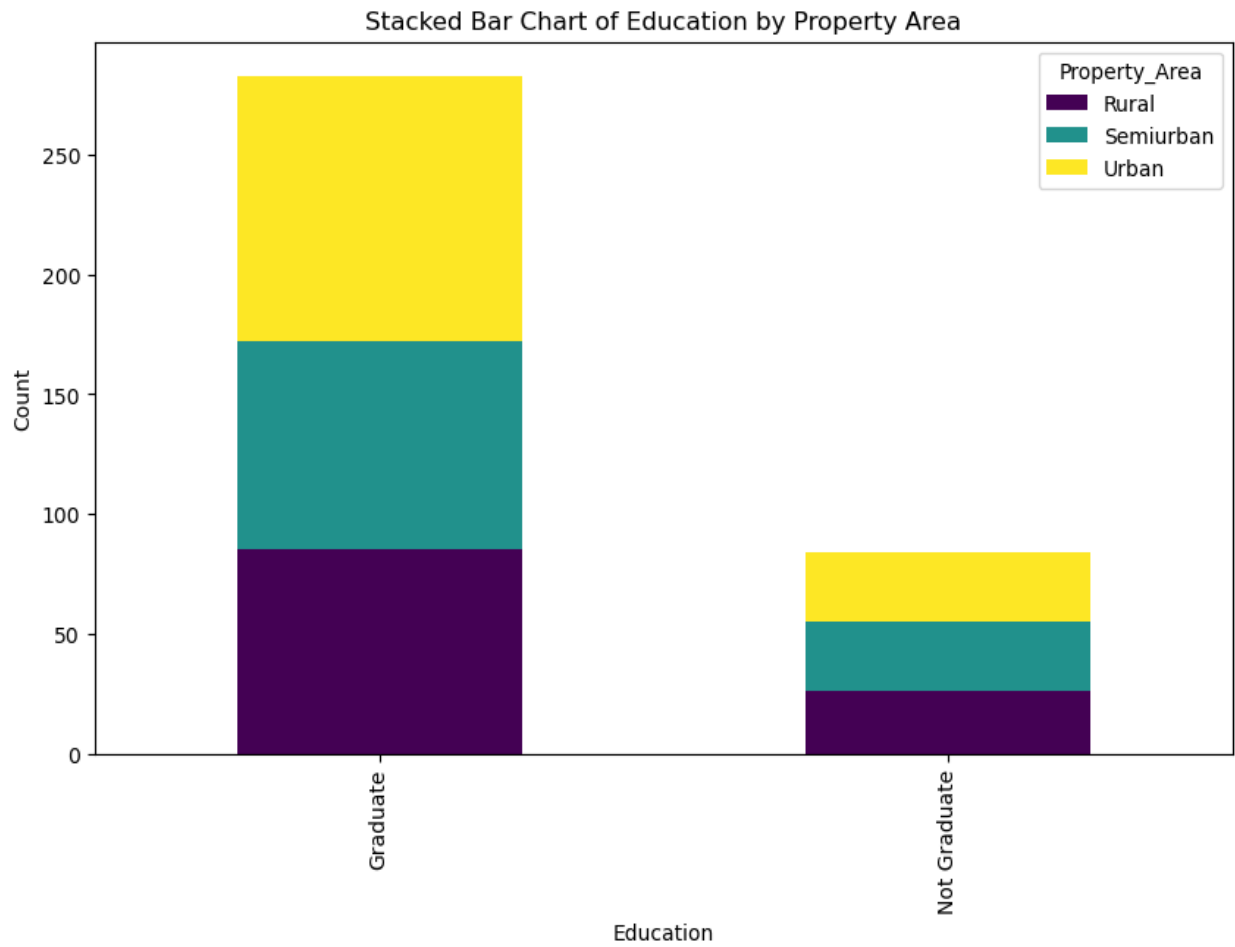
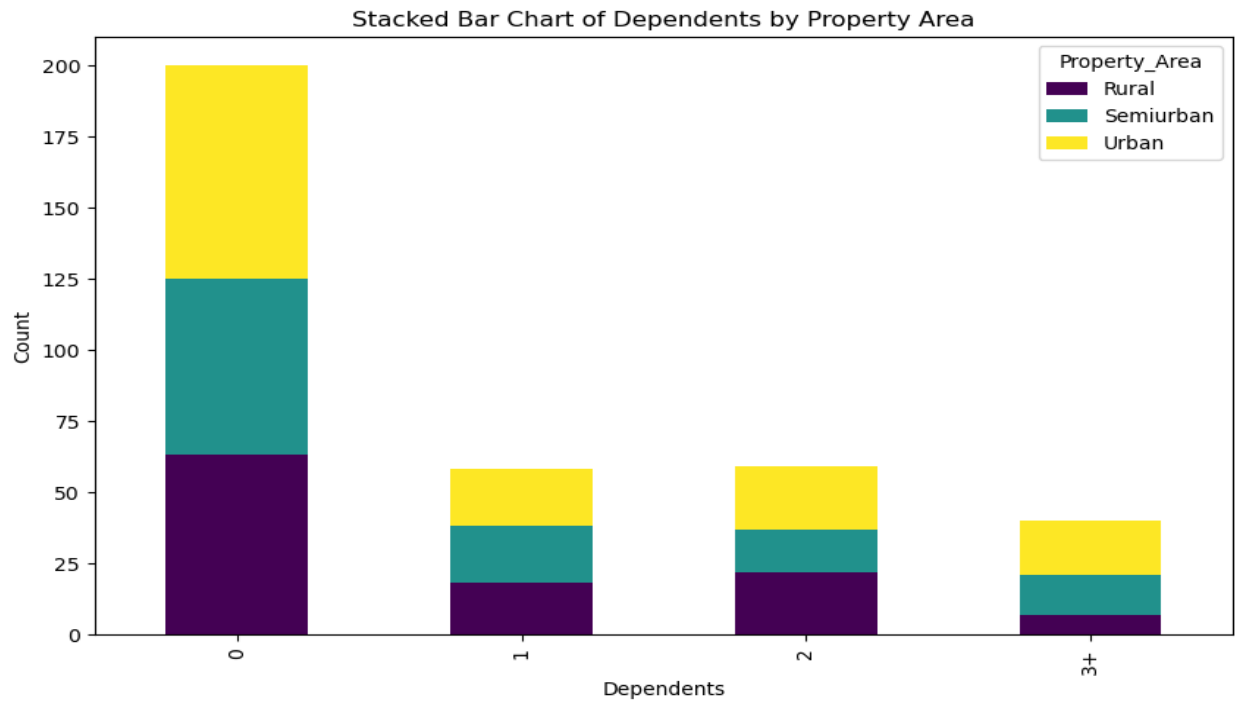
Perform a correlation analysis to identify relationships between numeric variables.

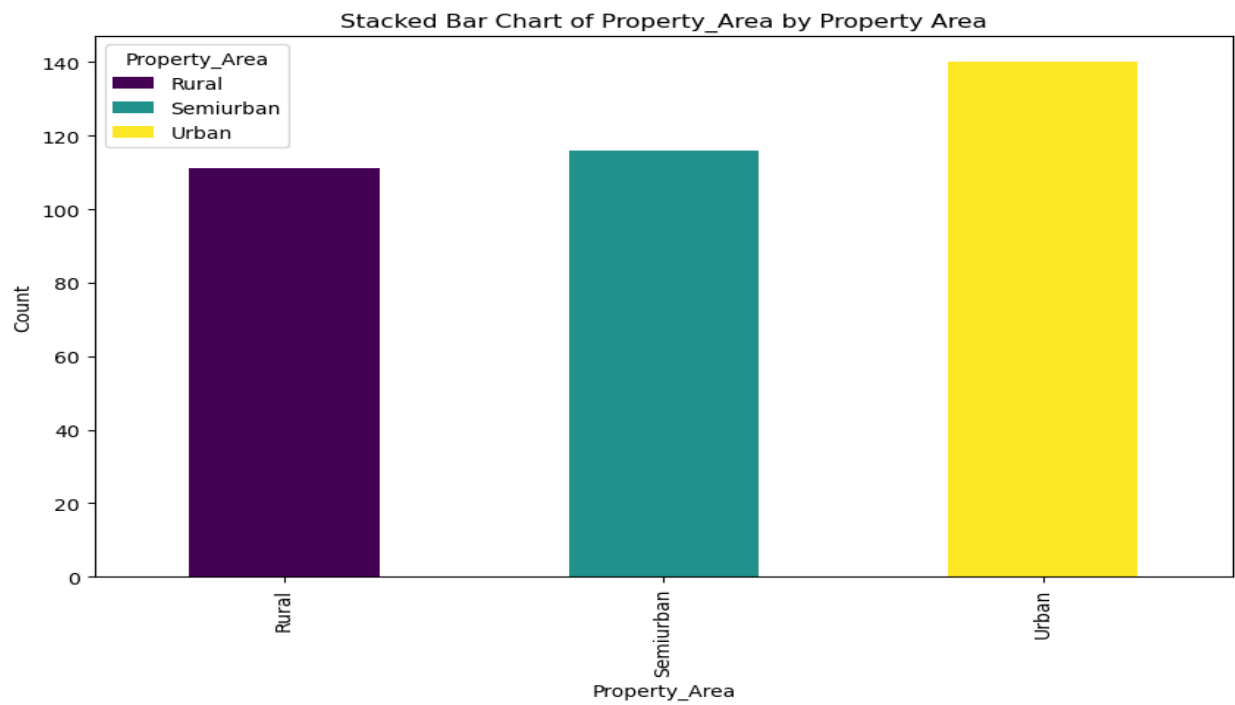
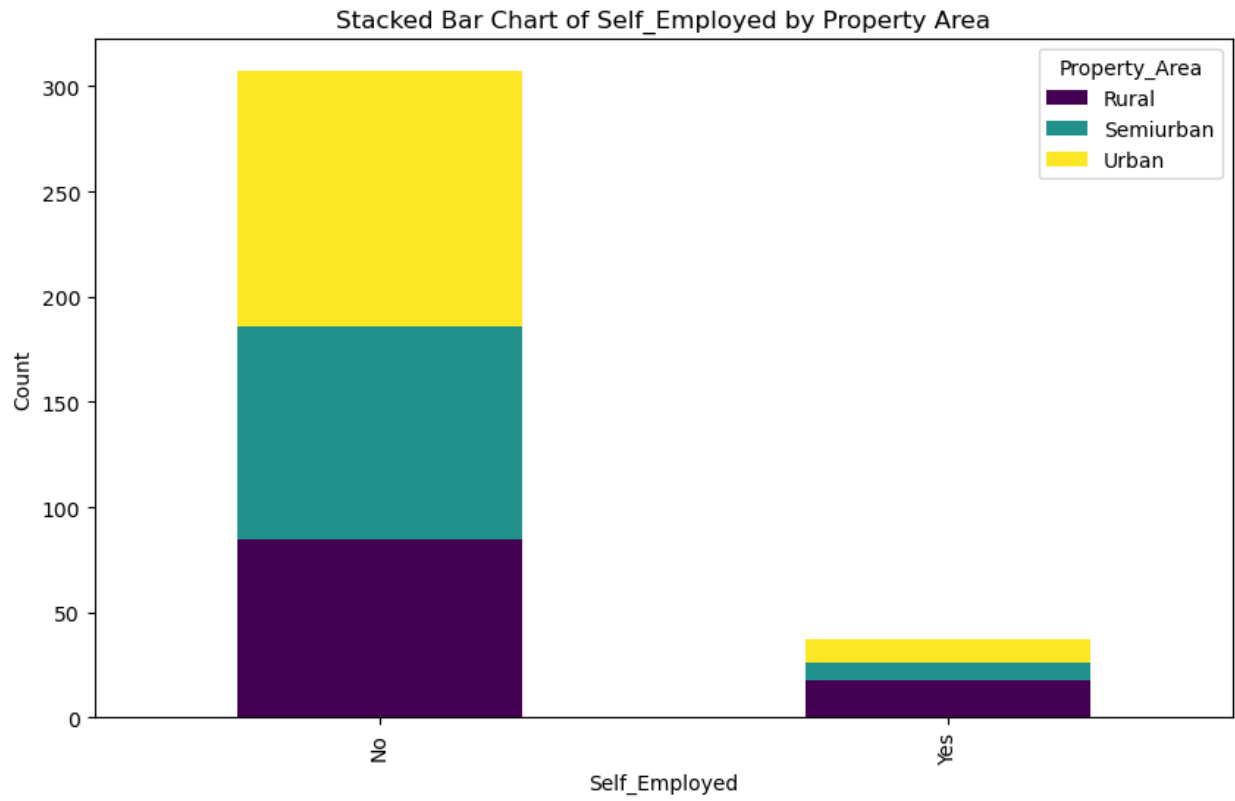
. heatmap



.Stacked barchart:Create a stacked bar chart to show the distribution of categorical variables across multiple categories.







Thank you

<https://colab.research.google.com/drive/18BqXSCqon69fvvzEuorjvWiQTr4Kavm7?usp=sharing>